

GENOMELENS

Public whole-genome demonstration

HG002 · Genome in a Bottle

Sanitized public demonstration generated from an end-to-end GenomeLens analysis of publicly released Genome in a Bottle data.

This public report contains a selected allowlist of results from the analysis. It is not the production GenomeLens report and does not expose the private analysis engine.

DATASET PROVENANCE

SOURCE Genome in a Bottle	REFERENCE BUILD GRCh38	BENCHMARK RELEASE v4.2.1	BENCHMARK SCOPE chr1-22
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EXTERNAL BENCHMARK

CONCORDANT 169/169	EXPLICIT TRUTH CALLS 169	MISMATCHES 0	NORMALIZATION FAILURES 0
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Concordance is measured on positions where the truth set states a genotype, inside the defined confident regions. It is not whole-genome accuracy, not clinical validation, and not an endorsement by NIST or GIAB.

SELECTED GENOMIC RESULTS

FINDING	CATEGORY	DISPOSITION	EVIDENCE BASIS
PTPN22 R620W carrier — autoimmune screening awareness	carrier	unresolved	curated clinical evidence
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)	pharmacogenomics	conditional	published pharmacogenomic guideline evidence
Elevated type-2-diabetes polygenic risk	polygenic risk	conditional	published association evidence

HEALTH-ECONOMIC RESULTS

REFERENCE-CASE NMB \$957	CANONICAL EXPECTED NMB \$1,239	INCREMENTAL COST \$367	INCREMENTAL QALYS 0.0132
ICER 27,734	NOTE Reference-case and canonical pathway results answer different questions and are never summed.		

SCOPE AND LIMITATIONS

- Public demonstration output; not clinical validation.
- Benchmark scope is limited to the stated GIAB panel and chr1-22.
- Modeled economics are not realized savings.
- Representative findings only; the full sanitized report is the complete output.

Methods and validation: [VALIDATION.md](#) · [ECONOMICS.md](#) · [LIMITATIONS.md](#)

Research software. Not a diagnostic device. Not medical advice.